

Intrinsic AKI — ATN, AIN, TMA & Crystal Nephropathy



1. ATN section — tubular injury

WHAT YOU SEE

ATN shelf banner

WHAT IT ENCODES

Acute tubular necrosis is the most common intrinsic AKI, from ischemia or nephrotoxins. Urine shows muddy-brown granular casts, FENa >2%, and renal tubular epithelial cells. Recovers over days-weeks if injury stops.

BOARD TRAP

ATN gives FENa >2% and muddy-brown casts; prerenal azotemia gives FENa <1% with bland sediment — key discriminator.

2. Gentamicin (aminoglycoside)

WHAT YOU SEE

Soldier holding gentamicin jar

WHAT IT ENCODES

Aminoglycosides cause non-oliguric ATN, typically after 5-7 days, dose-dependent, with hypomagnesemia. Accumulate in proximal tubule cells. Once-daily dosing reduces toxicity.

BOARD TRAP

Aminoglycoside ATN is classically NON-oliguric and delayed (~day 5-7) — not immediate; magnesium wasting is a clue.

3. Cisplatin

WHAT YOU SEE

Cisplatin jar / knight

WHAT IT ENCODES

Cisplatin is a direct proximal-tubule toxin causing ATN with magnesium and potassium wasting. Prevent with vigorous saline hydration and amifostine; carries dose-dependent cumulative nephrotoxicity.

BOARD TRAP

Cisplatin nephrotoxicity is prevented by aggressive IV hydration — magnesium wasting is the telltale electrolyte sign.

4. Ciprofloxacin / contrast (ATN context)

WHAT YOU SEE

Ciprofloxacin jar mid-shelf

WHAT IT ENCODES

Many drugs straddle ATN and AIN; iodinated contrast classically causes ischemic/toxic ATN with early FENa <1% and rapid recovery in 3-5 days. Risk highest in CKD, diabetes, volume depletion.

BOARD TRAP

Contrast 'nephropathy' often has a LOW FENa early (vasoconstrictive) — can mimic prerenal, unlike typical ATN.

5. Ibuprofen — NOT ATN

WHAT YOU SEE

IBUPROFEN X / NOT ATN placard

WHAT IT ENCODES

NSAIDs cause AKI mainly by afferent arteriolar vasoconstriction (prerenal, prostaglandin inhibition) and can cause AIN or minimal-change disease — not primarily direct tubular necrosis.

BOARD TRAP

NSAID AKI is prerenal/AIN, NOT classic ATN — placing ibuprofen in the ATN bin is the trap the X warns against.

6. AIN section — interstitial

WHAT YOU SEE

AIN shelf banner / sheriff

WHAT IT ENCODES

Acute interstitial nephritis is a hypersensitivity reaction: fever, rash, eosinophilia, eosinophiluria, WBC casts, and AKI days-weeks after a culprit drug. Treat by stopping the drug; steroids if severe/persistent.

BOARD TRAP

The classic AIN triad (fever, rash, eosinophilia) is present in a MINORITY of cases — its absence does not exclude AIN.

7. Ciprofloxacin / fluoroquinolone (AIN)

WHAT YOU SEE

Sheriff with ciprofloxacin

WHAT IT ENCODES

Fluoroquinolones, beta-lactams (especially methicillin/penicillins), and sulfonamides are classic AIN triggers via T-cell-mediated hypersensitivity. Onset is idiosyncratic, not dose-dependent.

BOARD TRAP

AIN is hypersensitivity (not dose-related) — rechallenge can recur faster; don't blame the dose.

8. Checkpoint inhibitor

WHAT YOU SEE

Checkpoint inhibitor monk figure

WHAT IT ENCODES

Immune checkpoint inhibitors (PD-1/PD-L1, CTLA-4) cause immune-mediated AIN, often weeks-months in, and frequently respond to corticosteroids. A growing oncology-era cause of AKI.

BOARD TRAP

Checkpoint-inhibitor AIN may need steroids AND drug hold — unlike simple drug AIN where withdrawal alone often suffices.

9. PPI / proton-pump inhibitor

WHAT YOU SEE

AIN drug class on shelf

WHAT IT ENCODES

Proton-pump inhibitors are now a leading cause of drug-induced AIN, often with insidious onset and minimal classic triad. Can progress to chronic interstitial nephritis if unrecognized.

BOARD TRAP

PPIs are an under-recognized, common AIN cause — easy to miss because they lack the fever/rash/eosinophilia triad.

10. NSAID → AIN

WHAT YOU SEE

NSAID fire extinguisher / AIN

WHAT IT ENCODES

NSAIDs can cause AIN that is often accompanied by nephrotic-range proteinuria from concurrent minimal-change disease, classically WITHOUT eosinophilia.

BOARD TRAP

NSAID-induced AIN typically lacks eosinophilia and may show nephrotic proteinuria — atypical for usual AIN.

11. Rifampin

WHAT YOU SEE

Rifampin jar

WHAT IT ENCODES

Rifampin causes AIN, sometimes with intermittent (interrupted) dosing and an immune-allergic mechanism that can include hemolysis and flu-like symptoms.

BOARD TRAP

Rifampin AIN is linked to intermittent dosing/re-exposure — continuous therapy is less likely to trigger it.

12. TMA section

WHAT YOU SEE

TMA shelf banner / spiders / webs

WHAT IT ENCODES

Thrombotic microangiopathy injures glomerular endothelium, producing microangiopathic hemolytic anemia (schistocytes), low platelets, and AKI. Causes include HUS, complement-mediated TMA, malignant hypertension, drugs.

BOARD TRAP

TMA shows schistocytes with a NORMAL coagulation profile (normal PT/PTT/fibrinogen) — distinguishing it from DIC.

13. Anti-VEGF / glomerular endothelium

WHAT YOU SEE

ANTI-VEGF web / glomerular endothelium label

WHAT IT ENCODES

Anti-VEGF agents (bevacizumab) and calcineurin inhibitors injure glomerular endothelium causing drug-induced TMA with hypertension and proteinuria. VEGF is needed for endothelial/podocyte health.

BOARD TRAP

Anti-VEGF damages endothelium (TMA + HTN + proteinuria) — not a tubular or interstitial process.

14. Crystal section — acyclovir IV

WHAT YOU SEE

CRYSTAL shelf / acyclovir IV / icicles

WHAT IT ENCODES

Crystal nephropathy: IV acyclovir (especially rapid bolus in volume-depleted patients) precipitates intratubular crystals causing obstructive AKI. Prevent with slow infusion and hydration.

BOARD TRAP

Acyclovir crystal AKI is prevented by hydration and SLOW infusion — rapid IV bolus in a dry patient is the setup.

15. Other crystal causes

WHAT YOU SEE

Crystal shelf jars / sulfa

WHAT IT ENCODES

Other crystal nephropathies: sulfonamides, methotrexate, indinavir, and tumor lysis (uric acid/calcium phosphate). Urine may show characteristic crystals; volume + sometimes urinary alkalinization help.

BOARD TRAP

Tumor lysis causes uric acid AND calcium-phosphate crystal AKI — rasburicase plus hydration, avoid alkalinization if phosphate high.

16. Phosphate binders only / patiromer

WHAT YOU SEE

Old woman: phosphate binders only, PATIROMER, X on sodium phosphate

WHAT IT ENCODES

Oral sodium phosphate bowel preps cause acute phosphate nephropathy (calcium-phosphate deposition) — avoid in CKD/elderly. Patiromer/sodium polystyrene treat hyperkalemia, not AKI.

BOARD TRAP

Sodium phosphate purgatives are nephrotoxic in at-risk patients — the X warns against them; use PEG-based preps instead.